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Key Points:

- We employed a state-of-the-art model to calculate the structural connectivity index across High Mountain Asia
- Structural connectivity influences sediment transport, with greater variability linked to increased sediment fluxes
- Spatial variations in structural connectivity are highly correlated with cryospheric landscape changes

Supporting Information:

Supporting Information may be found in the online version of this article.

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Structural Connectivity Shaped by Cryospheric Degradation Modulates Sediment Transport Dynamics in High Mountain Asia

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Abstract Rapid atmospheric warming has transformed cryospheric landscapes, affecting sediment transport processes. However, our understanding of hillslope and riverine connectivity in large headwaters remains limited. Here, we apply the Index of Connectivity (IC) to model structural connectivity for High Mountain Asia (HMA). Significant spatial heterogeneity in structural connectivity is observed across hillslope, river network, and basin scales associated with cryospheric changes. Structural connectivity regulates the transfer of sediment from hillslope erosion to river channels ($\rho = 0.77$, $P < 0.01$) and controls downstream sediment flux at the basin scale ($\rho = 0.79$, $P < 0.01$). We also found that climate change has driven geomorphic processes in cryospheric landscapes, thereby modulating the spatiotemporal variability of structural connectivity. Rising temperatures and altered precipitation patterns have intensified sediment supply and transport pathways, increasing cryospheric hazard risks in high-IC mountainous areas like the Himalayas-Hengduan ranges. Our findings underscore the importance of structural connectivity in understanding sediment transport dynamics in cold regions.

Plain Language Summary Rapid warming in High Mountain Asia is reshaping landscapes covered by permafrost, ice, and snow, changing how sediment moves from hillslopes to river systems. Using a computer model, we analyzed how well-connected different parts of the landscape are to allow sediment to travel. This landscape connectedness varies greatly and plays a key role in regulating sediment movement from mountain slopes, through river networks, and across entire basins. We found that areas with stronger physical links tend to have higher sediment transport downstream. Climate change has intensified processes like glacier retreat, permafrost thaw, and erosion, which have increased the supply and transport of sediment. These changes increase the risk of hazards in highly connected mountainous regions, such as the Himalayas and Hengduan ranges. Our study highlights how important these landscape connections are for understanding sediment dynamics in cold, high-altitude regions.

1. Introduction

In expansive riverine systems, connectivity has emerged as a critical element for understanding the underlying processes and intricacies of these systems (Cavalli et al., 2019). Over the last decade, the concept of structural (or “static”) connectivity has garnered increasing interest within the earth sciences community, and has been recognized as a cornerstone in the exploration of hydro-geomorphic processes (Fryirs, 2013; Harvey, 2001; Messenzehl et al., 2014; Najafi et al., 2021; Ortíz-Rodríguez et al., 2019). In mountainous basins, the spatial characteristics of structural connectivity are instrumental in deciphering the complexities of soil erosion, sediment availability, and the intricacies of sediment transport both within and between active geomorphological land-systems, notably slopes and fluvial networks (L. J. Bracken et al., 2013; L. J. Bracken et al., 2015; Heckmann et al., 2018). These dynamics are significantly influenced by persistent climatic change, tectonic activity, and transient human interventions (S. N. Lane et al., 2009; Mayor et al., 2008), markedly shaping the evolution of landscape patterns. Despite its critical significance, studies on water and sediment connectivity, as well as its impacts on sediment transport in cryospheric basins, remain limited. This is particularly notable in regions that are highly sensitive to modern climate change (East & Sankey, 2020; Hilger & Beylich, 2019).

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High Mountain Asia (HMA), encompassing the Tibetan Plateau and surrounding massifs, contains the headwaters of numerous major Asian rivers. The region is integral for its provision of vital ecological functions, hydropower potential, and natural resources such as soil, minerals, and biomass, underpinning the sustenance of nearly 2 billion individuals globally (Immerzeel et al., 2020; D. Li et al., 2022; J. Li et al., 2024; Yao et al., 2012). As the world's highest region and occupying the mid-latitudes, HMA is subject to a warming trend nearly twice the global mean, advancing at 0.32°C per decade (Mountain Research Initiative EDW Working Group, 2015). Consequently, HMA's river systems are increasingly sensitive and vulnerable to intricate tectonic dynamics and climate-induced ecological alterations. These include intensified temperature-dependent erosion (D. Li, Overeem et al., 2021; J. Li et al., 2023), shifts in sediment transport mechanisms (Shi et al., 2018; H. Wang et al., 2011), and a series of cascading hydro-geomorphic consequences (East & Sankey, 2020; J. Li et al., 2024). Contemporary research in HMA and its sub-basins consistently indicates a notable yearly increase in sediment fluxes (Figure 1) (Ali & De Boer, 2007; D. Li et al., 2020; D. Li, Lu et al., 2021; Wasson et al., 2022; F. Zhang et al., 2022), which are nearly double that observed in other cryospheric basins (e.g., Pan-Arctic regions and Greenland, Figure 1b). Moreover, the diminished connectivity in several river channels, coupled with increased sediment sinks—such as the expansion of thermokarst lakes and proglacial lakes, and the proliferation of thermal gullies—leads to intensified sediment deposition in upstream river reaches. This localized deposition contributes to a significant underestimation of basin-wide erosion rates when assessed solely through downstream in situ observations (J. Li et al., 2024).

Cryosphere degradation, including glacier retreat, permafrost thaw, and diminishing snow cover, significantly impacts sediment yield, transport, and deposition in HMA. These changes occur primarily through alterations to the nature and distribution of sediment sources and sinks, as well as by reshaping connectivity within and between hillslopes and fluvial systems (T. Zhang et al., 2022). For example, coupled ice margins occur where glaciers maintain direct hydrological connections with proglacial rivers, facilitating efficient meltwater and sediment transfer (Evans et al., 2020). In contrast, decoupled ice margins emerge as glaciers retreat and lose direct connectivity with these systems, leading to disrupted sediment and water flux pathways. These transitions are influenced by spatial and temporal changes driven by climate warming, which alter the hydrological and sediment dynamics of the system (Norris et al., 2024). In HMA's riverine network, structural connectivity exemplifies the integration of lateral (from slopes to channels) and longitudinal (from upstream to downstream) links. This connectivity integrates diverse cryospheric and alpine landscape elements, including glaciers, permafrost, proglacial lakes, and fluvial systems, facilitating sediment and water transfer across the landscape (Heckmann et al., 2018; J. Zhang et al., 2023). For example, recent findings indicate that glacial retreat facilitates the upward extension of meltwater-induced gullies, diminishing dependence on supra- and subglacial transport, thus expanding the contributing areas, and enhancing sediment connectivity and transport (S. N. Lane et al., 2017; Mancini & Lane, 2020). Additionally, permafrost and sediment collapse into evolving thaw pits within glacier ice contribute to the proliferation of thermokarst landscapes, producing active sediment sources and new erosion paths (D. Li, Overeem et al., 2021; Rudy et al., 2017). When sediment transport was impeded by sediment sinks in river channels and well-developed vegetation on hillslopes, sediment transport dynamics became dominated by deposition at these specific locations. However, effectively integrating the rapid transformations in sediment transport with connectivity poses a significant challenge, obstructing a comprehensive understanding of the influence of sediment connectivity changes on sediment transport processes (East & Sankey, 2020; Hooke & Souza, 2021; T. Zhang et al., 2022). We still lack a comprehensive understanding of the changes in sediment transport within the cryosphere and their driving mechanisms, as well as their subsequent impacts on biodiversity, water quality, and river health within HMA's river systems (Batista et al., 2022; Immerzeel et al., 2020).

Here, we endeavor to address two critical scientific gaps: how does climate change influence the connectivity of cryospheric landscapes, and how does the structural connectivity regulate the sediment transport processes in headwater regions? We employ a 30 m Digital Elevation Model (DEM) to model the structural connectivity index (IC) for almost 3.7 billion hillslope points across HMA. We analyzed the IC metrics of 496 sub-basins (HydroSHEDS Basins Level six, Hybas-6) in HMA (Figure 1), correlating them with meteorological and landscape indices. We aim to evaluate the influence of climatic and cryospheric changes on structural connectivity and to infer how connectivity impacts erosion and deposition patterns in HMA. This work is positioned at the forefront of unraveling these complex dynamics, offering a nuanced understanding of sediment transport in cryospheric environments under the evolving climate paradigm. Our findings not only advance our knowledge of

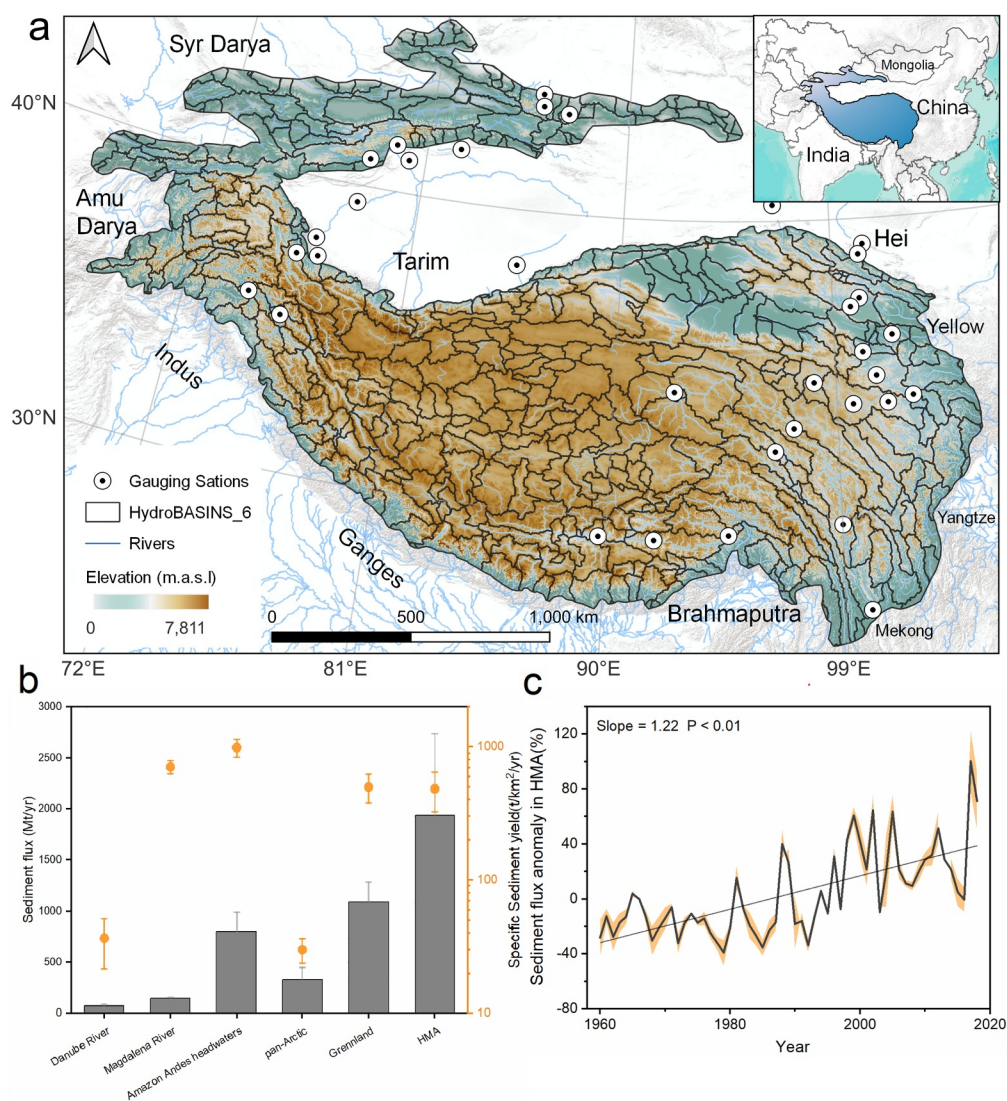


Figure 1. A basic, comprehensive overview of the geography and sediment variations in the headwater basins of the High Mountain Asia (HMA). (a) The geographical location, elevation, and distribution of sediment monitoring stations in HMA. The boundaries of the sub-basins are delineated based on the Hybas-6 products (Lehner & Grill, 2013), with the base map and inset provided courtesy of ESRI and USGS. (b) A comparison of contemporary sediment fluxes (Mt/yr) and specific sediment yield ($\text{t}/\text{km}^2/\text{yr}$) across the pan-Arctic (11.48 million km^2), Greenland (2.17 million km^2), HMA (4.72 million km^2), as well as several mid-latitude cryospheric river basins, including the Danube River, Magdalena River, and Amazon Andes headwaters. This analysis spans data collected from the 1960s to the 2020s and underscores the significantly higher sediment flux in HMA compared to other cryospheric basins (Li et al., 2021a; Overeem et al., 2017; Restrepo & Kjerfve, 2000; Syvitski, 2002). (c) The annual rate of change in sediment flux anomalies in HMA. Sediment flux anomaly denotes the relative deviation of sediment discharge data for a given year from the multi-year average value. It underscores a marked acceleration in sediment transport variations since the 21st century.

hydro-geomorphic processes but also provide valuable new insights into the mobilization, transport, and delivery of water and sediment across the interconnected slopes, gullies, and river channels in cold regions.

2. Materials and Methods

2.1. Calculating Structural Connectivity Index

Traditional methods for calculating structural connectivity are limited by computational resources and technological capabilities, posing significant temporal and spatial constraints for large-scale regions ($>10,000 \text{ km}^2$)

(Hooke & Souza, 2021; S. N. Lane et al., 2017). For instance, Heckmann et al. (2018) reviewed existing structural connectivity calculations and found that they are predominantly applied to small catchments ranging from 10^{-6} to 10^3 km², with few applications at larger scales. Current structural connectivity calculations are primarily conducted on ArcGIS platforms, which face limitations in computational resources when handling large-scale alpine basins. These methods are characterized by prolonged iteration times, cumbersome workflows, and an inability to process batch operations efficiently. Moreover, existing large-scale structural connectivity calculations often rely on coarse-resolution DEM data, resulting in spatial resolutions of structural connectivity outputs exceeding 500 m. Moreover, current understanding is predominantly confined to local scales—plots, hillslopes, and small basins—in various global contexts (Hooke & Souza, 2021; Messenzehl et al., 2014).

We utilized the state-of-the-art global DEM released by the National Aeronautics and Space Administration in 2020 (NASADEM), with a spatial resolution of 30 m. The NASADEM derives its foundation from advancements in processing techniques, elevation control, void-filling, and merging with data previously unavailable during the original Shuttle Radar Topography Mission production phase. Subsequently, all computations were conducted for HydroSHEDS Basins Level six (Hybas-6) basins, totaling 496 in the HMA region, with an average area of 28,690 km². In addition, we utilized the HMA 8 m DEM data set to analyze the temporal changes in IC within specific regions. However, it is important to note that this data set provides DEMs only for glacierized and snow-covered areas within HMA, and the derived products are not annual averages. As a result, we matched the DEM data with the corresponding climate and sediment variables based on the closest available dates, aiming to ensure temporal consistency in the analysis.

We first clipped the input DEM files using the Python platform to match the corresponding basin boundaries (Hybas-6) and performed sink-filling on the resulting sub-basin DEM files. The sink-filled DEM files were then iteratively processed using the TauDEM tool to generate D8 flow directions and accumulated flow. By setting a threshold for accumulated flow, we determined the initiation points of the stream network and subsequently generated stream network data. Additionally, water body data for each sub-basin was extracted from the Hybas-6 hierarchical basin data set and matched to the corresponding basin IDs. The buffered basin and water body data were then merged and saved as new shapefile files.

Each output shapefile was processed iteratively using GeoPandas' clip function to clip the water body data to the basin boundaries, and the clipped data was saved to a specified directory. For each sub-basin DEM file, we extracted basic raster information, replaced invalid data in the DEM with NaN, and used NumPy functions to create filters and processing matrices. Based on the sink_flag_tg function, we selected appropriate methods for handling different flow paths, calculated D8 and Dinf flow direction arrays, and assigned NoData values where applicable (Husic & Michalek, 2022; Tarboton, 1997). Using the computed D8 and Dinf flow direction arrays, we reclassified the D8 slope file and wrote the updated data into new GeoTIFF files. These files contained calculations of slope, relative smoothness index, and accumulated flow for each pixel, as well as the creation and computation of weighted flow length. Through iterative processing, we computed the IC, a metric delineating the potential connectivity between sediment sources and sinks. Widely employed in studies on sediment connectivity (Borselli et al., 2008), the IC is expressed as:

$$IC = \lg \frac{D_{up}}{D_{dn}} \quad (1)$$

The IC signifies higher connectivity as it increases, and lower connectivity as it decreases. Cavalli et al. (2013) utilized the D-infinity algorithm and restricted slope values to a range of 0.005–1.000 to refine the IC model. These modifications enhanced the simulation of optimal flow paths for connectivity and improved the interpretation of sedimentary geomorphic processes. The IC for sediment is composed of two components: an upslope component (D_{up}) and a downslope component (D_{dn}). Here, D_{up} denotes the potential for sediment transport downstream:

$$D_{up} = \bar{W} \bar{S} \sqrt{A} \quad (2)$$

Where A represents the watershed area; $\bar{S}$ denotes the average slope of the watershed; $\bar{W}$ signifies a weighted coefficient that comprehensively reflects the surface conditions and land use of the watershed (Equations 4 and 5). D_{dn} reflects the potential for sediment to be transported to the nearest sediment sink:

$$D_{dn} = \sum \frac{d_i}{W_i S_i} \quad (3)$$

Where d_i represents the distance of the i th cell to the sediment sink; W_i denotes a weighted coefficient that comprehensively reflects the surface conditions and land use of the i th cell; S_i stands for the slope of the i th cell. To prevent errors, we set that S_i less than 0.005 must be assigned a value of 0.005, while S_i greater than 1 must be assigned a value of 1.

The weighted coefficient W_i was originally derived from the C-factor in the Revised Universal Soil Loss Equation (RUSLE) framework, designed to evaluate basins characterized by agriculture and vegetation. It was later revised to incorporate the surface roughness index (R_i) for sediment connectivity studies in mountainous areas. To enhance the applicability of the IC model across diverse research contexts, Persichillo et al. (2018) introduced Manning's roughness values as weighting factors. For overland flow, the resistance associated with each land-use type is represented by the Manning's roughness coefficient (n) for surface flow. These values can be estimated without hydraulic parameters using empirical tables corresponding to specific land-use categories. Following Liu et al. (2024), we assigned Manning's n roughness coefficient to each land-use category within the HMA. Notably, glaciers and permafrost were incorporated during the assignment process. However, snow cover was excluded as it does not represent the underlying terrain surface and temporarily traps sediments (Mishra et al., 2019).

Essentially, IC expresses the structural connectivity of landscape units within the basins. However, after adjustment by the weighted coefficients, it also reflects functional connectivity to a certain extent (Wu et al., 2023). When revising the weighted coefficient W_i using random roughness (standard deviation of elevation in a 5×5 moving window), a negative correlation exists between random roughness and connectivity index, hence, standardization should be carried out according to the following equation (Equation 4). For each land-use category, we incorporated Manning's roughness coefficients (n) to reflect their influence on the weighting factor. Notably, the n values were assigned as follows: cropland (0.3), frost/shrub (0.4), grassland (0.25), water (0.1), barren (0.05), impervious (0.04), and wetland (0.2). Additionally, Manning's n values for glaciers were assigned as 0.03 for clean ice surfaces and 0.06 for debris-covered glacier surfaces. For permafrost, n values were set at 0.10 for bare, exposed permafrost and 0.15 for permafrost covered with sparse vegetation.

$$W_i = 1 - \frac{R_i}{R_{\max}} - n \quad (4)$$

Where R_i represents the random roughness; $R_{\max}$ denotes the maximum random roughness; and n represents the Manning's roughness coefficients. It can be expressed as:

$$R_i = 1 - \sqrt{\frac{1}{25} \sum_{j=1}^{25} (x_j - \bar{x})^2} \quad (5)$$

Where x_j is the elevation of the j th cell within the moving window and $\bar{x}$ is the mean elevation in the moving window.

2.2. Divided IC Into Two Components

To better reflect the influence of IC on sediment transport, we also computed two modified components of IC, namely IC_{channel} and IC_{outlet} . The main distinction between these two components lies in their target definitions and calculation methodologies. IC_{channel} initially defines the target for the upslope contributing area (e.g., the main river channel). It calculates the probability of sediment eroding from hillslopes reaching the nearest sink, thereby delineating hillslope-to-channel connectivity (Harvey, 2001). We employed the RUSLE to compute the average soil erosion modulus generated by water erosion across the entire HMA from 1992 to 2021 (Equation 6

and Text S1 in Supporting Information S1). The erosion modulus of hillslopes is coupled with $IC_{channel}$ to ascertain whether structural connectivity controls the process of sediment transport from hillslopes to river channels. Note that due to differences in national/regional policies and the resolution of publicly available data, we computed the erosion modulus of high-mountain ranges within China instead of the entire HMA. While the selected ranges share a broadly similar tectonic setting within the active Himalayan-Tibetan orogenic system, climatic regimes vary significantly (Yao et al., 2012). For instance, the Hengduan Mountains experience a monsoon-influenced climate, whereas the Tianshan Mountains are characterized by arid, continental conditions (D. Li et al., 2024). The equation is formulated as:

$$A = R L S K C P \quad (6)$$

where A is the predicted actual annual soil erosion ($t\ ha^{-1}\ yr^{-1}$) on land with vegetation cover and erosion control practices, R is the rainfall erosivity factor ($MJ\ mm\ ha^{-1}\ h^{-1}\ a^{-1}$), LS is the topographic factor (dimensionless) with L being the slope length factor and S being the slope factor, K is the soil erodibility factor ($t\ h\ MJ^{-1}\ mm^{-1}$), C is the vegetation cover and management factor (dimensionless), and P is the support practice factor (dimensionless). Each factor was computed based on the native resolution of the input data (e.g., DEM data at $30\ m \times 30\ m$, soil properties data at $250\ m \times 250\ m$). Subsequently, all factors were resampled to a uniform resolution using bilinear interpolation and then multiplied together to derive the soil erosion rate for each year. More details can be found in Text S1 in Supporting Information S1.

On the other hand, IC_{outlet} sets its target at the outlet of each sub-basin. It computes the probability of sediment reaching the outlet for each cell in the basin, defining upstream-downstream channel interactions or basin connectivity (Mishra et al., 2019). We compared the basin-wide sediment flux to IC_{outlet} for each sub-basin, spanning from 1986 to 2021, to assess whether IC regulates sediment transport across the entire river network. The sediment flux data set was developed by J. Li et al. (2024), and we calculated the annual average sediment flux for each basin outlet within the corresponding sub-basin boundaries.

Overall, $IC_{channel}$ emphasizes connectivity from hillslopes to channels, while IC_{outlet} underscores connectivity from within the basin to its outlet. By retaining the highest percentiles of flow accumulation areas determined by the TauDEM D8 algorithm, we established flow targets for each Hybas-6 basin using the IC model (Husic & Michalek, 2022). Some large lakes and thaw subsidence regions were also considered sediment sinks (Figure 2b and Figure S1 in Supporting Information S1). For each sub-basin, we conducted visual interpretation and manual verification to identify and correct errors, especially in arid regions like the Qaidam and Inner basins. In total, ~ 3.7 billion IC computations were performed to compute the two structural connectivity indices at a 30 m resolution for the HMA (4.72 million km^2).

2.3. Coupling With Climatic, Geomorphic, and Cryospheric Factors

We employed the Spearman correlation coefficient to investigate the relationship between climatic, geomorphic, and cryospheric variables and IC in each sub-basin, using P-values to evaluate statistical significance. We utilized available in situ observational data to the greatest extent possible to calculate the climatic and geomorphic variables required for our analysis (D. Li et al., 2024; D. Li, Lu et al., 2021). We supplemented other public data sets to examine changes in IC and their impact on river sediment transport. These supplementary data include variables associated with IC changes, such as temperature (Zhao et al., 2021), precipitation (Kirschbaum et al., 2020), runoff (L. Wang et al., 2021), permafrost, and glaciers (Yao et al., 2012). Additionally, we utilized the European Center for Medium-Range Weather Forecasts (ECMWF) fifth-generation reanalysis data set (ERA5) as a supplementary source for meteorological data in regions lacking ground-based observations. To ensure consistency, we aligned the temporal resolutions of supplementary data sets with sediment and IC data sets, ensuring one-to-one correspondence throughout the study period (1986–2021). Specifically, to handle the potential errors associated with integrating in situ and supplementary data, we employed the variance-upscaling Quantile Mapping method developed by Ma et al. (2024). This method was used to construct regional gridded climate data (e.g., precipitation and air temperature), from which we then calculated the sub-basin averages included in our correlation analysis.

For the primary spatial analysis, we adopted Hybas-6 as the unit of analysis to perform basin-averaged calculations. Specifically, for IC and its drivers, we computed basin-averaged values across the Hybas-6 units to

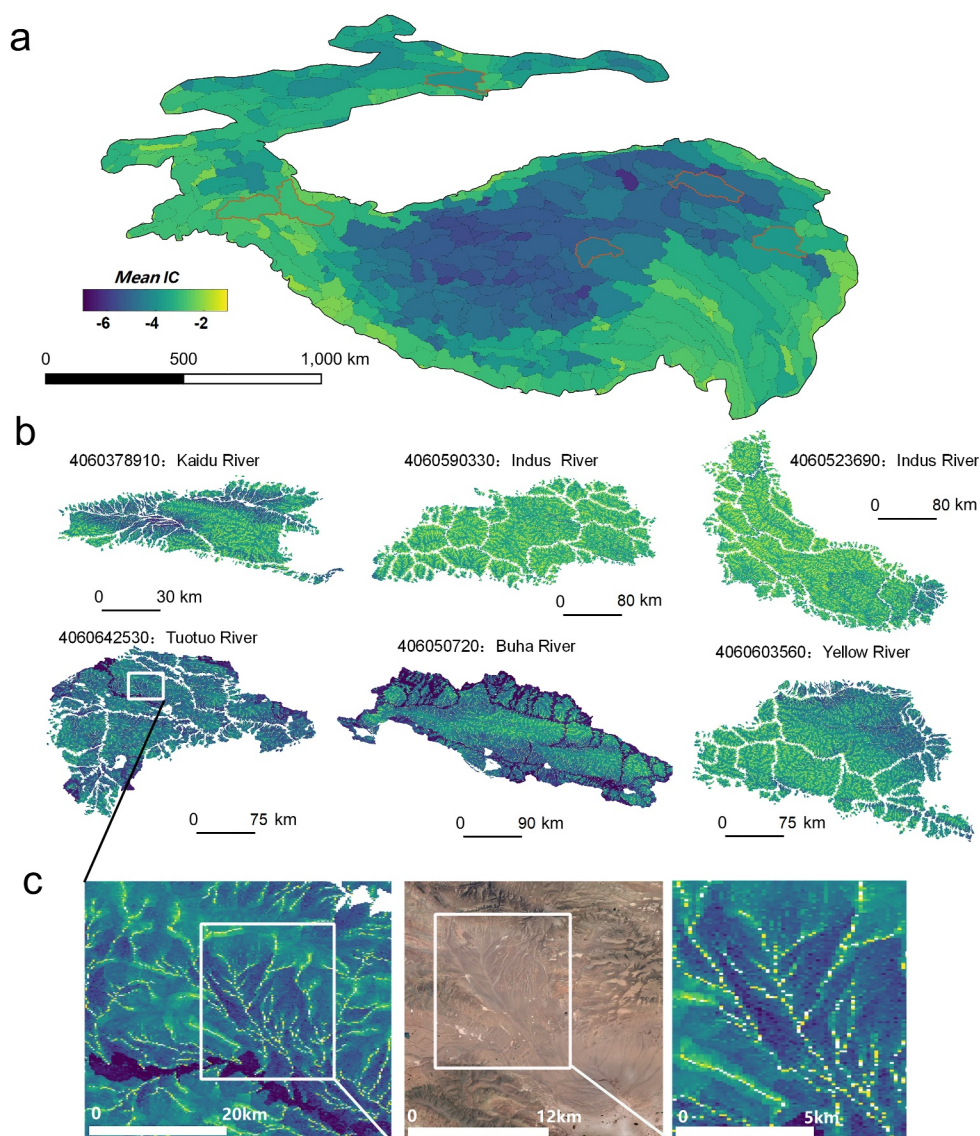


Figure 2. Spatial variability in the Index of Connectivity (IC) across different scales within the High Mountain Asia region, encompassing whole sub-basins and intra-basin hillslope-to-river channel dynamics. (a) Mean IC for all basins defined at the HydroSHEDS Basins Level six. (b) Detailed view of six representative basins from (a), illustrating the spatial variability in IC values. (c) Intra-basin variability in IC among hillslope surfaces, thermal gullies, and river channels, highlighting that despite the overall lower IC in the Tuotuohe sub-basin, areas within active erodible thermokarst landscapes exhibit significantly higher IC values compared to hillslope surfaces. The central point of the analyzed hillslope is located at 34.56°N, 91.36°E. Satellite imagery in panel (c) is sourced from the base map and inset, provided courtesy of the USGS.

facilitate our statistical analyses. The climatic variables analyzed for each sub-basin include air temperature, saturated water vapor pressure, aridity index, soil moisture content, precipitation, and evapotranspiration for each sub-basin. In terms of tectonics or geomorphology, we compared IC with elevation, slope, and the Terrain Ruggedness Index. Note that the variables used for statistical analysis are not identical to those used in the IC calculations. For instance, the DEM utilized in IC computation is processed with sink-filling to ensure hydrological continuity, whereas the DEM used for statistical analysis remains unprocessed in its original form. Additionally, we examined the association between IC and glacier-related variables (such as glacier retreat length and glacier runoff) and land surface temperature (LST) in each sub-basin. The glacier elevation change data were sourced from Shen et al. (2022) and glacier runoff data were sourced from the H. Wang et al. (2022).

Table 1

Comparison of Basin-Averaged Index of Connectivity and Suspended Sediment Flux in the Headwater Regions of High Mountain Asia Rivers

Headwaters	Syr darya	Chu-talas R.	Tianshan M.	Yili R.	Manasi R.	Amu darya	Indus	Tarim	Ganges
IC	−0.49	−0.68	−0.36	−0.87	−3.39	−2.95	−3.20	−3.43	−3.06
SE	0.33	0.43	0.27	0.28	0.31	0.53	0.87	1.00	0.78
SSF(Mt/yr)	12.31	5.69	20.15	2.51	3.35	24.52	32.11	18.14	3.63
SE	2.87	1.34	8.10	0.39	2.79	3.17	5.86	2.28	0.49
Headwaters	Inner	Qaidam	Brahmaputra	Hexi	Yellow	Yangtze	Mekong	Salween	
IC	−4.81	−4.62	−3.44	−3.71	−3.86	−3.48	−3.06	−3.38	
SE	0.73	0.79	0.91	0.97	0.91	0.99	0.62	1.11	
SSF(Mt/yr)	0.89	5.10	8.38	1.19	4.53	8.30	5.02	9.15	
SE	0.24	1.00	1.24	0.35	0.73	1.58	1.04	1.45	

Note. IC refers to IC_{out} in this context, and SSF represents the annual mean sediment flux monitored from the 1960s to the 2010s downstream. For headwaters without sediment monitoring at the outlet, we supplemented the data using the data set provided by J. Li et al. (2024).

3. Results and Discussion

3.1. Structural Connectivity in HMA Across Scales

We observed pronounced spatial heterogeneity in the structural connectivity across all HMA, characterized by a distinct “concave pattern” with lower connectivity in central areas and higher along the peripheries (Figure 2, Table 1). Approximately half of the sub-basins exhibit relatively low connectivity, with the lowest IC values concentrated in the Qaidam (-6.15 ± 1.26 , Mean $\pm$ SE) and the central Inner basin (-5.90 ± 1.36 , Figure 2a). These regions feature relatively flat terrain, and are marked by an arid climate with minimal precipitation, leading to extensive coverage of salt lakes and marshes in low-lying areas (L. Wang et al., 2020; Yao et al., 2012). For example, the Qaidam Basin, one of Earth's driest non-polar regions, is characterized by intermittent rivers constituting the majority (>90%) of its river systems (X. Li et al., 2023), and the presence of over 20 saline lakes (H. Li et al., 2019). Additionally, IC values across 120 sub-basins in the Inner Basin range between -5.90 ± 1.36 and -3.12 ± 1.05 . The flat terrain, coupled with the development of large tectonic depositional basins and the dry climate, significantly contributes to the lower structural connectivity in these areas. Existing studies further reveal that the accumulated sediment depths in these regions can reach between 10 and 14 km, underscoring that reduced connectivity is a major factor leading to widespread sediment deposition (Bao et al., 2017; F. Zhang et al., 2022).

The basins along the Himalayas and the Hengduan Mountains exhibit higher IC values (Figure 2a), encompassing the Amu Darya (-2.95 ± 0.53), Indus (-3.20 ± 0.87), Ganges (-3.06 ± 0.78), Brahmaputra (-3.44 ± 0.91), Mekong (-3.06 ± 0.62), and Salween rivers basins (-3.38 ± 1.11). The elevated IC in these basins can be attributed to a combination of climatic and tectonic influences (East & Sankey, 2020). From a climatic perspective, these river basins are subject to dynamic and intensive hydrological cycles (Hilger & Beylich, 2019; Immerzeel et al., 2020). Climate-driven landscape changes, including the melting of glaciers and snowpacks, and permafrost thawing, have facilitated the development of new flow paths and the expansion of active erodible landscape (Figure S1 in Supporting Information S1). For example, the proliferation of supraglacial streams, englacial channels, gully erosion (also known as thermal erosion gullies (TEGs), and thaw slumps have enhanced the connectivity in these regions (D. Li, Overeem et al., 2021; T. Zhang et al., 2022). On the tectonic front, the Himalayan region is one of the most seismically active areas in the world, due to the ongoing collision between the Indian and Eurasian plates (Ding et al., 2022). The long-term tectonic activity of the region has resulted in considerable uplift and landscape reshaping, setting the stage for high rates of erosion and sediment transport. While the direct impact of tectonic uplift within the past decades is negligible, the actively evolving topography influenced by these long-term processes contributes to the observed patterns of erosion and sediment dynamics (Z. Li et al., 2020). We found that the frequent seismically induced landslides contribute significantly to the sediment fluxes in these rivers. However, when landslide-activated sediment inputs become so substantial, they can overwhelm the transport system, creating new upstream sediment sinks and thereby decreasing IC values. The

combination of high relief, steep slopes, and active land surface processes facilitates a more efficient transfer of sediment from the hillslopes to the river channels, further contributing to the heightened connectivity in these basins (Figure 2a). Additionally, moderate connectivity is observed in basins concentrated in the Tianshan region (northern part of HMA) (-3.82 ± 1.15) and the upstream regions of the Yangtze (-3.48 ± 1.08) and Yellow River sources (-3.86 ± 1.14) in the east. These areas are characterized by higher connectivity near river segments, whereas connectivity decreases along mountain ridges and in locations distant from hydrological convergence points (Figure 2b).

The spatial heterogeneity of the IC is evident not only across different basins but also within individual basins where the overall IC is lower, yet regions of higher connectivity may exist. To illustrate this variability, we highlight six Hybas-6 basins. Multiple basins surrounding the Himalayas and the Hengduan Mountains, including the Indus, Ganges, Salween, Mekong, and Jinsha River basins, are identified as exhibiting relatively high connectivity (Figure 2). These basins are characterized by steep slopes and medium-grained, easily erodible soils throughout. For instance, the Indus River basin exhibits a generally high level of IC across its region (Figure 2b). Conversely, in basins with overall lower connectivity, such as the sub-basins of the source region of the Yangtze River, the Buha River, and the middle reaches of the upper Yellow River, extensive permafrost and seasonally frozen ground create micro-erosional units (e.g., thermal gullies, thaw slumps, and thermokarst depressions) during degradation processes. These units act as localized sediment sources and connectivity hotspots, despite the lower overall IC levels of these basins (Figures 2a and 2b, Figure S2 in Supporting Information S1). For example, in the Tuotuo River sub-basin on the northern slopes of the Tanggula Mountains (91.38°E , 34.54°N), the formation of TEGs enhances sediment connectivity between the river channels and more distant slope sediment sources, resulting in significantly higher IC in these gullies than on the slopes (Figure 2c) (Godin & Fortier, 2012; Rodzik et al., 2009). During thaw or rainy seasons, increased surface runoff transports sediment from slopes and gullies downstream. Additionally, when temperatures rise, the melting of exposed ground ice can trigger thaw slumps (Kokelj et al., 2013; D. Li et al., 2020). The occurrence of thaw slumps is highly temperature-sensitive, stabilizing below 0°C and reactivating above the freezing point. Similarly, in the Tuotuo River and within the aforementioned basins, numerous large channel-connected thaw slumps and thermal erosional landscapes, such as active-layer detachment, TEG, freeze-thaw erosion, thermal bank erosion, solifluction, and mudflows, are observed (D. Li et al., 2022; J. Li et al., 2023; Shi et al., 2018; T. Zhang et al., 2022) (Figure S1 in Supporting Information S1). These geomorphological formations contribute to an elevated IC and foster a relatively higher level of effective sediment transport in these specific locales.

3.2. The Primary Determinants of Structural Connectivity

In the HMA, climatic and tectonic forces predominantly account for the spatial variability observed in the IC across and within its river basins. Climate has long been recognized as a pivotal factor in shaping terrestrial landscapes, with the interplay of precipitation, runoff, and erosion, thus playing a central role in defining river connectivity (Ferrier et al., 2013; Hooke & Souza, 2021; Tucker & Slingerland, 1997). For example, a contemporary global analysis has identified that climatic aridity emerges as a first-order control on the concavity of river channels (Chen et al., 2019), exerting a significant impact on fluvial transport. However, other researchers posited that tectonic processes, rather than climatic factors, are the primary drivers of river basin evolution, thereby significantly impacting water and sediment flux within fluvial systems (Husic & Michalek, 2022; Michalek et al., 2023; Seybold et al., 2021).

We find that geomorphic factors, particularly tectonic activities, serve as the primary control on the structural connectivity within the HMA, exhibiting pronounced spatial coherence with IC across various scales (Figure 2 and Table S1 in Supporting Information S1). The rank correlation between geomorphic indices and IC is significantly stronger than that between connectivity and the six climatic indicators (Figure 3), underscoring that in the high-mountain regions, geomorphic effects, rather than climate, predominantly govern the spatial patterns of IC variability. Notably, in regions surrounding tectonic plate margins, where tectonic activity is heightened—such as in the sub-basins of the Hengduan Mountains and circum-Himalayan regions—river channels tend to exhibit more pronounced incision and higher sediment export (D. Li, Lu et al., 2021; Mishra et al., 2019). This phenomenon is attributed to the accelerated rates of regional tectonic uplift, culminating in a significant augmentation of the average IC within these basins. At the sub-basin scale, this phenomenon manifests as increased slope gradients and greater variations in topographic relief. When river channels are steep and sediment transport pathways are deeply incised, the connectivity between hill slopes and river networks is enhanced

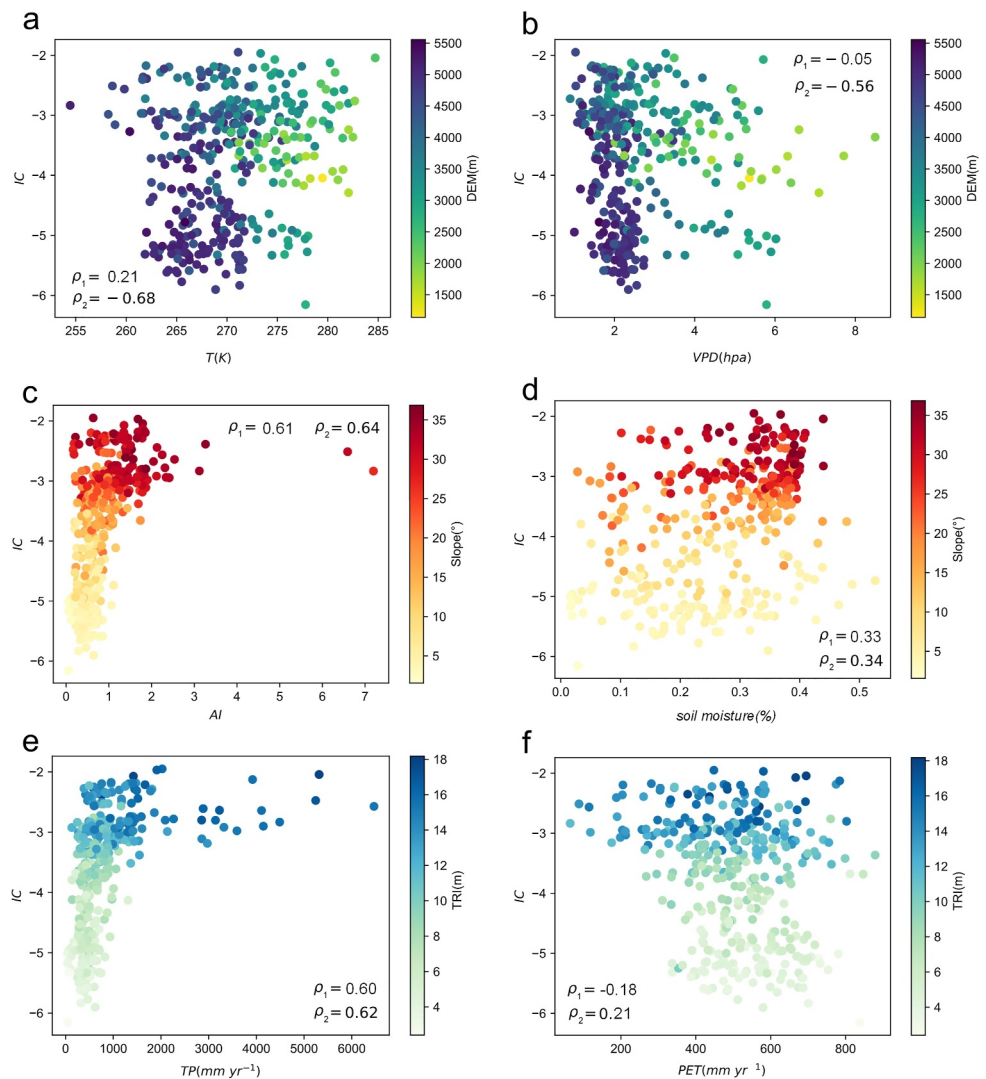


Figure 3. Correlations between Index of Connectivity (IC) and measures of geomorphic and climate at the Hybas-6 sub-basins. Climatic measures include (a) air temperature, (b) Atmospheric vapor pressure deficit, (c) aridity index, (d) soil moisture, (e) precipitation, and (f) potential evapotranspiration. Geomorphic measures include (a)–(b) elevation, (c)–(d) slope, and (e)–(f) terrain ruggedness index. The Spearman rank correlation (ρ) is shown; ρ_1 denotes the relationship between IC and climatic indices, while ρ_2 represents the relationship between IC and geomorphic indices.

(Figure 2b and Figure S2 in Supporting Information S1). This process is also strongly influenced by precipitation, which drives runoff and facilitates the mobilization and transfer of sediment into fluvial systems (D. Li et al., 2024).

Regarding climatic factors, spatial coherence is observed between the IC and various climatic drivers, albeit with notable heterogeneity in their relationships. Specifically, temperature ($\rho = 0.21$) and soil moisture content ($\rho = 0.33$) exhibit a modest positive correlation with IC, suggesting that higher temperatures correlate with intensified surface activities (e.g., permafrost thaw, glacier retreat, and the formation of thermokarst features), leading to a concurrent elevation in IC. Precipitation ($\rho = 0.60$) shows a strong correlation, diverging from the weak correlations observed in non-cryospheric basins. For instance, in their investigation into the relationship between the IC and precipitation across the continental United States, Husic and Michalek (2022) discerned a correlation coefficient of -0.31 . Collectively, these correlations between IC and temperature and precipitation indicate that the HMA is significantly influenced by climatic changes, which are exacerbating water erosion and thermal erosion (via the melting of glaciers and snowpack, and permafrost thawing), thereby driving spatial changes in IC more acutely than in non-cryospheric regions globally. However, potential evapotranspiration

displays a weak negative correlation, aligning with global studies and suggesting lower structural connectivity in wetland regions, where substantial volumes of sediment are deposited (D. Li et al., 2022; J. Li et al., 2024).

3.3. Structural Connectivity Control Sediment Transport Process

We also found significant variances in the role of structural connectivity in controlling sediment transport processes across distinct spatial scales. The spatial variability of total sediment flux correlates positively with structural connectivity, particularly within larger basins ($\rho = 0.41$), although the relationship is not statistically significant (Figure S3 in Supporting Information S1). The dynamics of erosion and sediment transport within larger basins are further modulated by an array of factors including local climate, vegetation, soil and rock types, and underlying surface conditions such as glaciers, permafrost, barren lands, and vegetated areas (S. N. Lane et al., 2009; Mayor et al., 2008; Michalek et al., 2023) (Figure S4 in Supporting Information S1), with structural connectivity metrics failing to capture the real-time dynamics of sediment transport (T. Zhang et al., 2023). Conversely, at the sub-basin (Hybas-6) and other smaller scales, structural connectivity serves as a dominant indicator in driving sediment transport.

We computed two distinct definitions of structural connectivity. IC_{channel} calculates the likelihood of sediment eroded from hillslopes reaching the nearest sink, initially defined for the upslope contributing area (Mishra et al., 2019). In contrast, IC_{out} sets the target at the basin outlet, computing the IC_{out} for each pixel in the catchment through the upslope contributing area, subsequently deriving the area-weighted IC_{out} . By comparing IC_{channel} for 275 sub-basins with the average basin-scale sediment yield (1992–2021, see Methods), we found that structural connectivity effectively regulates the transfer of sediment from hillslopes to river channels ($\rho = 0.77$, $P < 0.01$, Figure 4c). Furthermore, at the scale of the entire HMA region, the sediment export from sub-basin outlets to downstream areas also exhibits a significant positive correlation with IC_{out} ($\rho = 0.79$, $P < 0.01$, Figure 4d). At the river segment scale, a strong coupling between IC and sediment transport processes is also evident ($\rho = 0.71$, $P < 0.001$, Figure 4b). These findings highlight the capacity of structural connectivity at large scales to characterize the transportability of sediment across hillslope gullies and river channels, despite prior studies of sediment transport responses spanning from $\sim 10 \text{ km}^2$ (Cavalli et al., 2013) to $\sim 50,000 \text{ km}^2$ within smaller basins (Mishra et al., 2019).

At smaller scales, IC embodies both longitudinal and lateral connectivity (Hooke & Souza, 2021); the former pertains to the downstream transfer of water, energy, sediment, nutrients, or biota along channels (IC_{out} ; Stuart N. Lane et al., 2017), while the latter refers to the cross-sectional interaction between slope-channel or channel-alluvial plain interfaces (IC_{channel} ; Heckmann et al., 2018). Our analysis across most sub-basins within the HMA region indicates that longitudinal connectivity significantly modulates sediment export, with IC_{out} responding consistently to topographical variations (Figure S5 in Supporting Information S1) and suspended sediment dynamics (Figures S7a and S7b in Supporting Information S1). For example, in terrains characterized by deep valleys (alpine valley rivers), large thermal gullies ($> 100 \text{ m}$), or paraglacial zones in the upper Yangtze, Brahmaputra, Salween, and Indus River basins, sharp increases in IC are observed (Figure 2b and Figures S2c and S5c in Supporting Information S1). This suggests that cryosphere degradation notably enhances sediment connectivity in the longitudinal dimension, with newly formed gullies and runoff pathways during intense rainfall or high meltwater periods bolstering slope-channel linkages, thereby augmenting sediment transport from source areas and significantly increasing overall basin sediment flux (Figure S6 in Supporting Information S1) (D. Li, Overeem et al., 2021; Mancini & Lane, 2020).

Furthermore, when considering lateral connectivity, micro-geomorphological features play a crucial role. For instance, thermokarst slumps, small thermal/erosion gullies, and alluvial fans promote significant IC fluctuations, facilitating sediment deposition (Figures S3b and S7c in Supporting Information S1). This lateral dynamic, particularly across varying landscapes and slopes, impedes the efficient downstream transfer of increased upstream sediment fluxes (Figure S5d in Supporting Information S1). Additionally, large saline and thermokarst lakes expand the potential sink area, enhancing lateral connectivity between surrounding hillslopes and river networks, yet their propensity for sediment trapping significantly diminishes downstream longitudinal connectivity (Kondolf et al., 2014; S. N. Lane et al., 2017). This phenomenon underpins the lower average IC and extensive sediment deposition observed in regions such as the Tuotuo River (the source region of the Yangtze River), the Yellow River, and the Brahmaputra River Basins, all experiencing a decline in sediment delivery ratios (Figure S3b in Supporting Information S1).

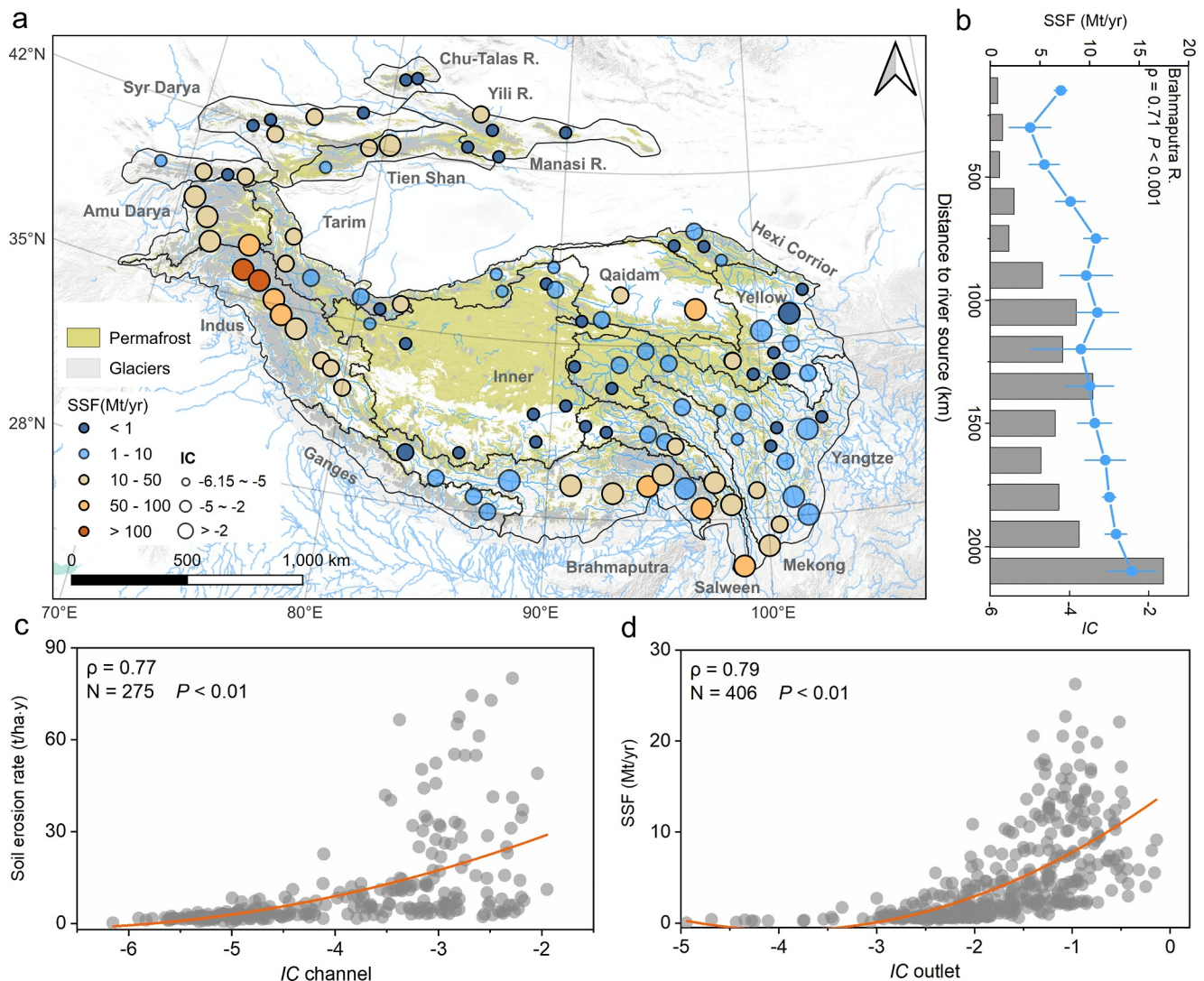


Figure 4. The regulation of sediment transport in the High Mountain Asia (HMA) by structural connectivity across multiple scales. (a) Spatial variation of sediment flux within the HMA over 36 years. We utilized sediment data extracted from the Tibetan Plateau riverine suspended sediment flux (SSF) data set formulated by J. Li et al. (2024). This data set has undergone area-weighted normalization to ensure consistency in runoff and sediment data representation. The spatial distribution of higher sediment flux coincides with areas characterized by elevated Index of Connectivity (IC). (b) The fluctuations in IC and its consequent impact on sediment transport along the course of the Brahmaputra River, with data synthesis performed on a 30 m resolution grid, providing insights into the fine-scale interactions between landscape connectivity and sediment mobilization. (c) The soil erosion yields of 275 sub-basins within the Qinghai-Tibet Plateau (QTP) exhibit a positive correlation with the variability of IC from hillslopes to river channels, indicating that IC governs the sediment transport processes from hillslope erosion to river channels. Note that due to differences in national/regional policies and the resolution of publicly available data sources, we computed the high-mountain ranges within China (QTP) instead of the entire HMA. (d) The statistically significant positive correlation between average IC and SSF among 406 sub-basins (in the HMA) classified under the HydroSHEDS Basins level 6 (Hybas-6) underscores the predictive value of IC in sediment transport processes at this granularity.

3.4. Structural Connectivity and Cryospheric Landscapes Change

Our investigation examines the interplay between structural connectivity and distinctive geomorphic alterations within the cryosphere, as well as the cascading impacts that these changes can trigger. We found a positive correlation between structural connectivity and cryospheric landscape changes (e.g., glacial retreat, permafrost disturbances, and snowpack melting), underscoring the efficacy of IC as a predictive tool for cryospheric and other hydro-geomorphic extremes (Husic & Michalek, 2022; Mishra et al., 2019). Over the past four decades, $\sim 29.92\%$ of the total sediment flux in the basins has been constrained (or deposited) within areas comprising only $13 \pm 2.7\%$ of the HMA, which emphasizes a sharp decline in local IC. We also found that climate change, through increasing temperature and altering precipitation patterns, has driven the dynamic evolution of cryospheric

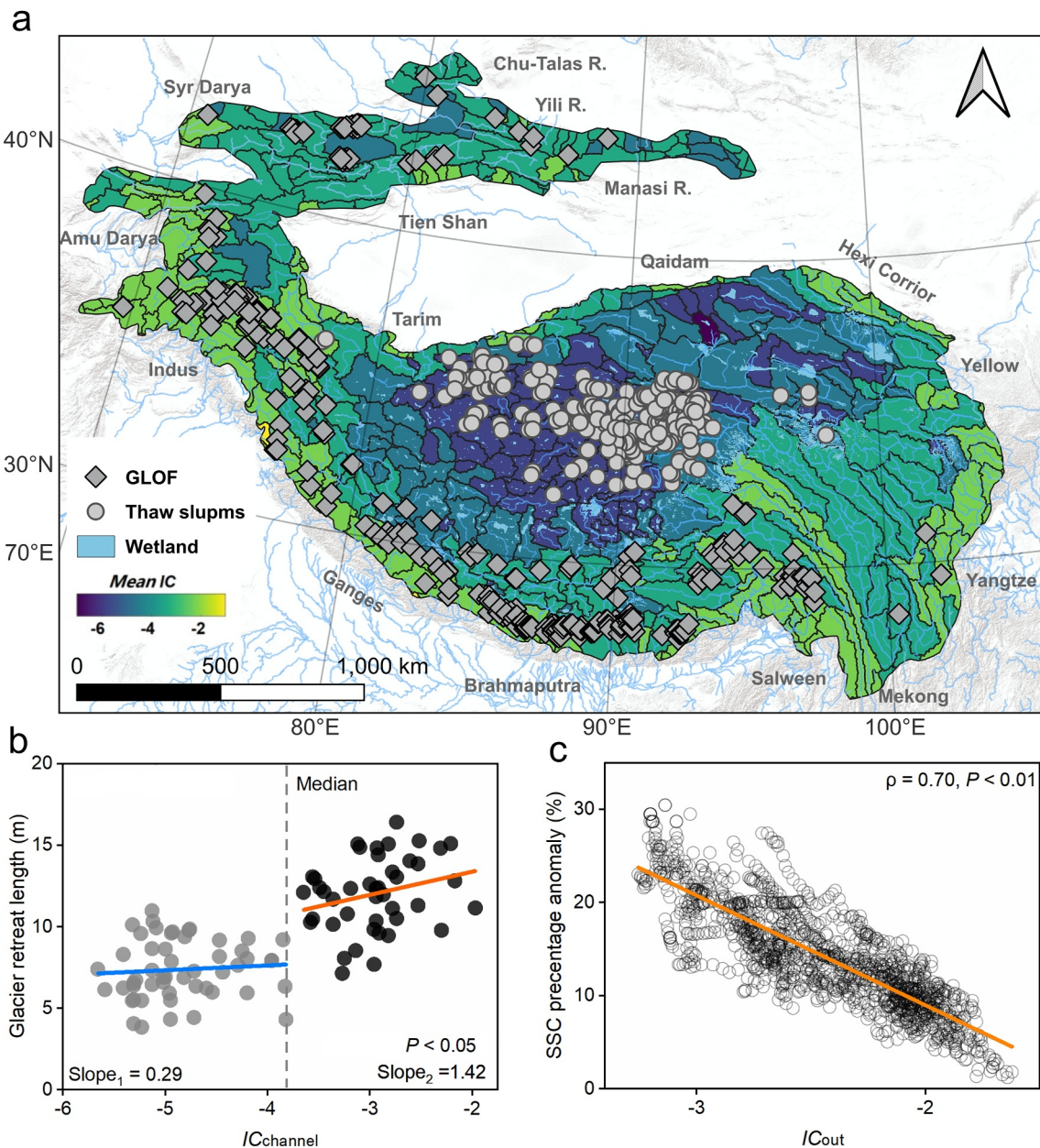


Figure 5. The response of structural connectivity to cryospheric landscape changes and associated hazards. (a) The spatial distribution Index of Connectivity (IC) highly correlates with landscape changes, with thaw slump regions and wetlands concentrated in regions with lower connectivity, while glacial lake outburst floods (GLOFs) concentrate in regions with higher connectivity along the Himalayan Mountain. The distributions of thaw slump data are from Luo et al. (2022), and GLOF data are from Tang et al. (2023), with the base map from USGS. (b) The relationship between the average retreat length of 95 glaciers in High Mountain Asia (HMA) over the past four decades and the regional $IC_{channel}$. Before reaching the median value of $IC_{channel}$, there is no apparent correlation between the two. However, when a certain IC threshold is surpassed, the impact of glacier retreat on sediment connectivity can be reflected by IC. Glacier elevation change data are from Shen et al. (2022). (c) Across the entire HMA region, the locations of suspended sediment concentration (SSC) anomalies exhibit a negative correlation with the IC_{out} at specific locations. SSC anomalies represent the percentage difference between the local SSC and the basin-weighted SSC. Typically, SSC anomalies indicate that sediment deposition occurs in the area.

components (Figures 6a and 6b). These changes have further influenced geomorphic processes, such as retrogressive thaw slump (RTS), riverbank erosion, and glacial lake expansion, thereby modulating the spatiotemporal variability of IC.

We quantified the relationship between the average retreat length of glaciers at specific 95 locations (from 1970s to 2010s) and the local average IC. We found that when glacier retreat surpasses a critical threshold—defined as a combination of reduced glacier length and thinning—and the IC value exceeds -3.85 , the local IC exhibits a

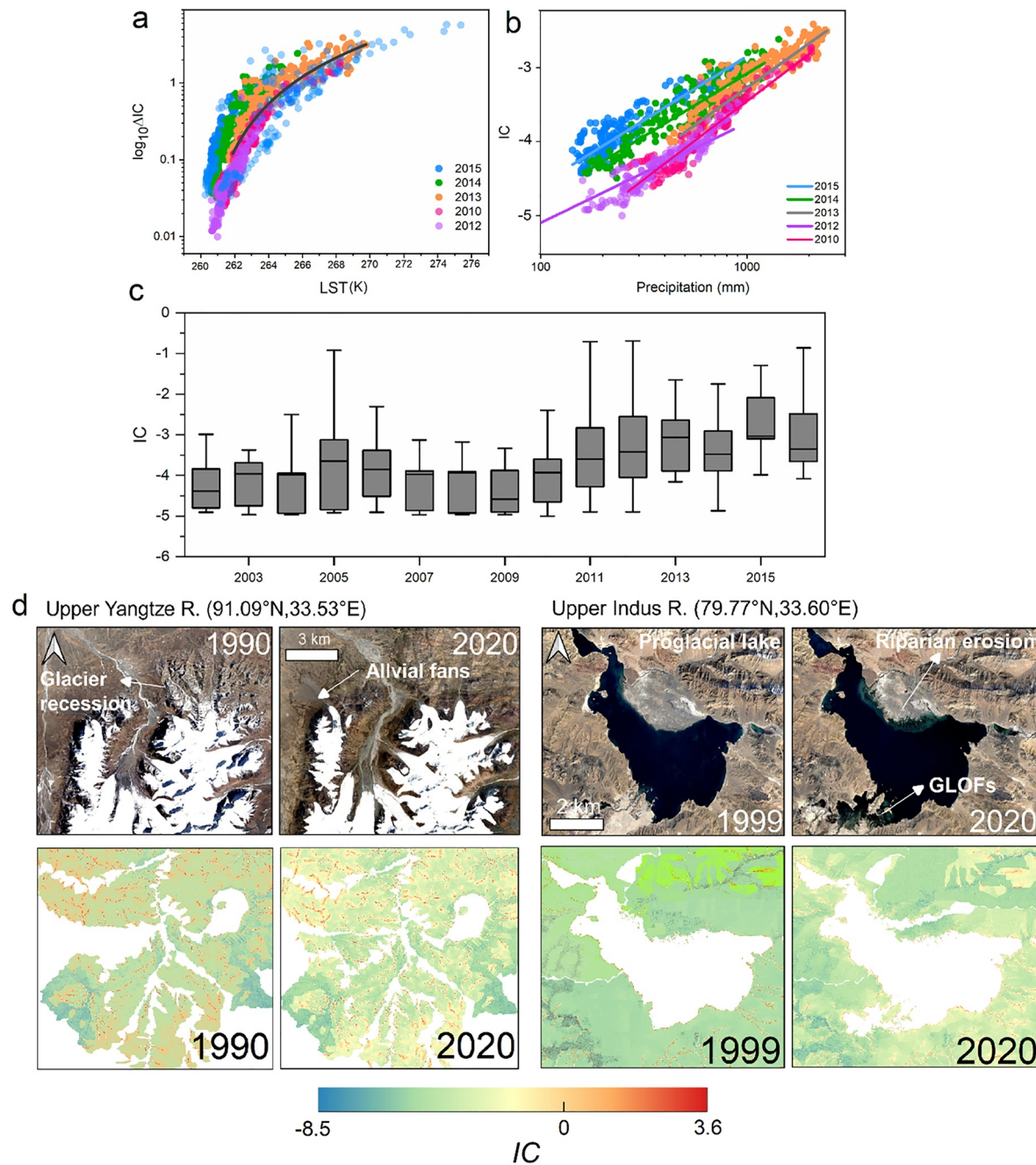


Figure 6. Impacts of climate change on the cryosphere and connectivity dynamics in High Mountain Asia (HMA). (a) Land surface temperature exhibits a positive logarithmic correlation with the regional average structural connectivity difference (ΔIC). The ΔIC values are normalized based on single-day Index of Connectivity (IC) and annual mean IC calculations. (b) IC demonstrates a positive correlation with precipitation during the period 2010–2015, with a slight interannual increase. (c) Temporal variations in IC are observed within glacier and snow-covered regions across HMA. (d) Remote sensing imagery comparisons for glaciers, lakes, and erosion gullies in the source regions of the Yangtze River and Indus River, along with IC comparisons. The observations reveal that glacier tongue retreat has promoted alluvial fan development, proglacial lake shoreline collapse and expansion, and snowmelt-induced permafrost gully erosion. Consequently, IC values have increased, with a notable rise in the number of high-connected gullies (red lines). Note that current medium-resolution and lower Digital Elevation Model (DEM) products (<30 m) in panel (d) are only available as single-scene data sets (1990, 1999, 2000, and 2020). The time-series IC calculations in this study were derived from the HMA 8 m DEM data set spanning the years 2002–2016.

positive correlation with the degree of glacier retreat ($P < 0.05$). Namely, once glacial basins surpass a critical IC threshold, substantial sediment stored within glacial systems becomes mobilized, facilitating its transport into river networks and enhancing downstream connectivity (McVey et al., 2023) (Figure 5b). Glacial recession and

thinning, with an average rate of -0.22 ± 0.05 m/yr across HMA, can expose surface crevasses and subglacial channels formed under various conditions, enhancing connectivity between hillslopes and proglacial areas (Hugonnet et al., 2021; Yao et al., 2012; T. Zhang et al., 2022). In these steep, tectonically active regions, substantial volumes of coarse-grained debris accumulate on glacier surfaces. This debris is subsequently transported down-ice and deposited as extensive moraine systems. The occurrence of ice margin decoupling determines whether the debris can be effectively transferred downstream (S. N. Lane et al., 2017). For instance, high alpine glaciers, particularly those in the Himalayas, often exhibit decoupled ice margins, which limit sediment transfer from glacial systems to proglacial environments, thereby reducing downstream sediment flux and altering sediment connectivity (Atkinson et al., 2016). However, once a certain threshold of sediment transport potential is surpassed—such as through the erosion of moraine belts that re-establish coupling between the ice margin and proglacial areas—sediment output from glacial meltwater begins to dominate downstream sediment dynamics (Huss, 2018; Livingstone et al., 2016). For example, at the Geladandong Glacier (the source region of the Yangtze River), the retreat of the glacier tongue has exposed additional sediment transport pathways (Figure 6d). This retreat has led to increased sediment erosion at the glacier's forefront, driving the development of proglacial alluvial fans. Consequently, regional IC has shown an upward trend over time (Figure 6d and Figure S9 in Supporting Information S1).

In permafrost-dominated basins, characterized by relatively flat terrains and lower IC, extensive lakes and wetlands are prevalent (Figure 5a). Spatial variability in connectivity within permafrost basins is significant, potentially disconnecting local sediment sinks from downstream areas and watershed outlets, including regions of RTS, thermokarst lakes, other water bodies, and lower-lying wetlands (Figure 5a and Figure S2 in Supporting Information S1). The percentage of annual suspended sediment concentration anomalies across the entire HMA region (1986–2021) exhibits a significant negative correlation with the overall basin-scale IC_{out} , further indicating the effectiveness of IC in reflecting whether sediment tends to be deposited locally or be transported downstream at the river segment scale (Figure 5c). These deposition events are observed in the Inner and Qaidam basins, as well as the upper reaches of the source regions of the Yangtze, Yellow, Salween, and Brahmaputra Rivers, where IC_{out} is significantly lower than in the middle and lower reaches (Figure 5a).

Climate change has also intensified glacial and permafrost landscape dynamics, affecting IC in temporal dimensions. Rising temperatures and changing precipitation regimes have accelerated geomorphic processes such as thermokarst development, riverbank erosion, and thaw slump expansion, thereby increasing local IC and sediment export (Figures 6a and 6b, and Figures S1 and S9 in Supporting Information S1). For example, while IC typically exhibits minimal temporal variability over short timescales (Figure 6b) (East & Sankey, 2020), we identified a positive interannual correlation between IC and both LST and precipitation, particularly after 2008/2009 (Figures 6a and 6b). These processes have restructured new sediment transport pathways, locally enhancing connectivity across multiple spatial scales, including hillslope-to-channel, channel-to-river network, and upstream-to-downstream linkages (Figure 6c). For instance, thaw slumps can release substantial sediment loads (Luo et al., 2022), transiently increasing downstream connectivity before sediment deposition stabilizes within wetlands or floodplains (Figures 6b and 6c). In basins with higher IC and greater topographic relief, these cryospheric landscapes' changes are expected to scale nonlinearly with warming and wetting (Kokelj et al., 2013; Lewkowicz & Way, 2019), driving disproportionate increases in sediment fluxes until the landscapes re-equilibrate (T. Zhang et al., 2023).

Furthermore, cryospheric disasters such as glacial lake outburst floods, landslide lake outburst floods, and river damming caused by ice avalanches, permafrost riverbank erosion, thawing mudflows and debris flows, are particularly prone to occur in the highly glaciated basins along the Himalayan regions (e.g., the Indus, Tarim, Ganges, and upper Brahmaputra basins, Figure 5a) (D. Li et al., 2022; M. Tang et al., 2023). These basins exhibit high rates of rock uplift and higher basin-weighted IC values, with significant increases in runoff and sediment transport observed over the past few decades (D. Li, Lu et al., 2021). Despite the rapid response of water and sediment to climate change, sediment tends to be deposited in areas with relatively low IC or at sites prone to cryospheric hazards (Figures 4a and 5a, and Figures S1 and S2 in Supporting Information S1). These sediment traps with lower IC (such as wetlands, lakes, well-developed braided channels, and vegetated shallow areas) have the capacity to accumulate vast quantities of sediment, which can be rapidly mobilized in the event of extreme rainfall, snowmelt, or dam failure, posing a significant threat to both local ecology and hydroelectric projects.

3.5. Limitations and Perspectives

To date, in the HMA, apart from sporadic and discontinuous hydrological and sediment records from 28 basin outlet hydrological stations (Figure 1) (D. Li, Lu et al., 2021), our exploration of the relationship between IC and water-sediment dynamics relies solely on limited and publicly available data (J. Li et al., 2023; Shi et al., 2018; F. Zhang et al., 2022). More extensive data sets would mitigate these issues and facilitate clearer connections between data sets. A limitation of our structural connectivity modeling approach is its focus solely on static surface features, neglecting potentially crucial vertical connectivity between river and groundwater (G. Wang et al., 2009). More precise and physics-based modeling of dynamic fluxes between surface, subsurface, and groundwater could enhance predictions of hydrological extremes, especially considering subglacial conduits capable of transporting significant sediment fluxes (T. Zhang et al., 2022). However, such vertical connectivity models necessitate substantial data and modeling resources, while the fundamental theories and observations regarding the energy exchange between surface and subsurface water remain lacking (Heckmann et al., 2018; Hooke & Souza, 2021). Moreover, while certain large thaw erosion gullies or thaw slumps can be detected using 30 m resolution DEM (Figure 6d and Figure S5 in Supporting Information S1) and satellite imagery (Ma et al., 2025; Yang et al., 2025), the changes and impacts of most subglacial channels, retrogressive thaw slumps, and other microtopographic features may be challenging to directly capture due to the limited resolution. However, even if individual features are of lower resolution (<30 m), their collective influence or the processes driving their formation can manifest as detectable changes in the IC patterns at sub-watershed scales and beyond.

In addition, there remains a lack of unified representation of functional connectivity (or dynamic connectivity) at a large regional scale. Functional connectivity delineates the interactions among surface processes within systems possessing certain levels of structural connectivity (e.g., runoff, sediment transport, pollutant movement), serving as a crucial indicator for quantifying hydrological and sediment connectivity (López-Vicente & Ben-Salem, 2019; McVey et al., 2023). The time scale of climate change may be too short to manifest in structural connectivity changes. For example, the relationship between IC and LST or precipitation, as calculated in Figures 6a and 6b, shows minimal interannual variability, with data points exhibiting a relatively clustered distribution. However, functional connectivity may respond to climate change, as it is driven by short-term climatic variables such as precipitation (Wu et al., 2023). However, several challenges hinder its broader application. Firstly, it has primarily been applied to small catchments ranging from 10^{-6} (Wohl et al., 2019) to 10^3 km² (Wu et al., 2023), and its scalability to larger extents remains unexplored due to computational demands. Sediment connectivity characterizes the degree to which sediment within a basin is transported by runoff, inevitably influencing soil erosion dynamics. Over certain time scales, such as the formation and development of rill networks, as well as the rapid development of gullies during warming periods, enhanced sediment connectivity provides convenient pathways for both surface runoff aggregation and sediment transport, exacerbating soil erosion (S. N. Lane et al., 2017). Conversely, soil erosion types, intensity, and their spatiotemporal distributions inevitably affect sediment connectivity, constituting a novel topic in soil conservation. Urgent efforts are needed to (a) develop functional connectivity models at regional and even global scales and/or (b) conduct studies examining the relationship between static indices and functional connectivity across various hydro-meteorological forcing scenarios. Such analyses will enhance our comprehension of the linkages between indices and observed (or modeled) system responses (Heckmann et al., 2018), a topic scarcely addressed to date.

4. Conclusions

In this study, we employed a state-of-the-art model to calculate the structural connectivity index for nearly 3.7 billion hillslope points across HMA. Our results revealed pronounced spatial heterogeneity in structural connectivity, characterized by a distinct “concave pattern” with lower connectivity in central regions and higher connectivity along the peripheries. Tectonics and geomorphology factors are primary controls that determine hydro-sedimentary connectivity at basin and sub-basin scales, respectively. Over the past four decades, ~29.92% of the total sediment flux in the basins has been constrained or deposited within areas comprising only $13 \pm 2.7\%$ of HMA, coinciding with a sharp decline in local IC. Structural connectivity describes the efficiency of sediment transfer from hillslope erosion to river channels and is closely linked to aggregate downstream sediment flux at the basin scale. Spatial variations in IC exhibit strong correlations with cryospheric landscape changes, such as glacier retreat, permafrost thaw, and wetland development. Climate change has driven the dynamic evolution of

cryospheric components, influencing geomorphic processes (including retrogressive thaw slumps, riverbank erosion, and the expansion of glacial lakes) and thereby modulating the spatiotemporal variability of IC within cryospheric landscapes. These findings emphasize the importance of structural connectivity as an indicator for understanding erosion and sediment transport dynamics in cold regions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All original data and codes used in this manuscript are available at the Zenodo data portal (<https://zenodo.org/records/10872458>) and are cited and referenced in the manuscript. The NASADEM data set is downloaded from the Laboratory (2021). Boundaries of glaciers and permafrost are obtained from Zhao et al. (2021). Land surface temperature data are available at W. Tang et al. (2024). The HMA 8 m DEM data are obtained from Dandabathula et al. (2024). HydroSHEDS Basins Level six (Hybas-6) products are obtained from Lehner and Grill (2013). The fifth-generation ECMWF Reanalysis (ERA5) data set is downloaded from Copernicus Climate Change Service (2019).

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References

- Ali, K. F., & De Boer, D. H. (2007). Spatial patterns and variation of suspended sediment yield in the upper Indus River basin, northern Pakistan. *Journal of Hydrology*, 334(3–4), 368–387. <https://doi.org/10.1016/j.jhydrol.2006.10.013>
- Atkinson, N., Pawley, S., & Utting, D. J. (2016). Flow-pattern evolution of the laurentide and cordilleran ice sheets across west-central alberta, Canada: Implications for ice sheet growth, retreat and dynamics during the last glacial cycle. *Journal of Quaternary Science*, 31(7), 753–768. <https://doi.org/10.1002/jqs.2901>
- Bao, J., Wang, Y., Song, C., Feng, Y., Hu, C., Zhong, S., & Yang, J. (2017). Cenozoic sediment flux in the qaidam basin, northern Tibetan Plateau, and implications with regional tectonics and climate. *Global and Planetary Change*, 155, 56–69. <https://doi.org/10.1016/j.gloplacha.2017.03.006>
- Batista, P. V. G., Fiener, P., Scheper, S., & Alewell, C. (2022). A conceptual-model-based sediment connectivity assessment for patchy agricultural catchments. *Hydrology and Earth System Sciences*, 26(14), 3753–3770. <https://doi.org/10.5194/hess-26-3753-2022>
- Borselli, L., Cassi, P., & Torri, D. (2008). Prolegomena to sediment and flow connectivity in the landscape: A GIS and field numerical assessment. *Catena (Cremlingen)*, 75(3), 268–277. <https://doi.org/10.1016/j.catena.2008.07.006>
- Bracken, L. J., Turnbull, L., Wainwright, J., & Bogaart, P. (2015). Sediment connectivity: A framework for understanding sediment transfer at multiple scales. *Earth Surface Processes and Landforms*, 40(2), 177–188. <https://doi.org/10.1002/esp.3635>
- Bracken, L. J., Wainwright, J., Ali, G. A., Tetzlaff, D., Smith, M. W., Reaney, S. M., & Roy, A. G. (2013). Concepts of hydrological connectivity: Research approaches, pathways and future agendas. *Earth-Science Reviews*, 119, 17–34. <https://doi.org/10.1016/j.earscirev.2013.02.001>
- Cavalli, M., Trevisani, S., Comiti, F., & Marchi, L. (2013). Geomorphometric assessment of spatial sediment connectivity in small alpine catchments. *Geomorphology*, 188, 31–41. <https://doi.org/10.1016/j.geomorph.2012.05.007>
- Cavalli, M., Vericat, D., & Pereira, P. (2019). Mapping water and sediment connectivity. *Science of the Total Environment*, 673, 763–767. <https://doi.org/10.1016/j.scitotenv.2019.04.071>
- Chen, S.-A., Michaelides, K., Grieve, S. W. D., & Singer, M. B. (2019). Aridity is expressed in river topography globally. *Nature*, 573(7775), 573–577. <https://doi.org/10.1038/s41586-019-1558-8>
- Copernicus Climate Change Service. (2019). ERA5-land hourly data from 1950 to present [Dataset]. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. <https://doi.org/10.24381/CDS.E2161BAC>
- Dandabathula, G., Roy, S., Ghatage, O. S., Kolase, V. B., Satpute, S., Ghosh, K., et al. (2024). Performance validation of High Mountain Asia 8-meter digital elevation model using ICESat-2 geolocated photons. *Journal of Mountain Science*, 21(8), 2562–2578. <https://doi.org/10.1007/s11629-024-8820-8>
- Ding, L., Kapp, P., Cai, F., Garzone, C. N., Xiong, Z., Wang, H., & Wang, C. (2022). Timing and mechanisms of Tibetan Plateau uplift. *Nature Reviews Earth and Environment*, 3(10), 652–667. <https://doi.org/10.1038/s43017-022-00318-4>
- East, A. E., & Sankey, J. B. (2020). Geomorphic and sedimentary effects of modern climate change: Current and anticipated future conditions in the western United States. *Reviews of Geophysics*, 58(4). <https://doi.org/10.1029/2019RG000692>
- Evans, D. J. A., Atkinson, N., & Phillips, E. (2020). Glacial geomorphology of the neutral hills uplands, southeast alberta, Canada: The process-form imprints of dynamic ice streams and surging ice lobes. *Geomorphology*, 350, 106910. <https://doi.org/10.1016/j.geomorph.2019.106910>
- Ferrier, K. L., Huppert, K. L., & Perron, J. T. (2013). Climatic control of bedrock river incision. *Nature*, 496(7444), 206–209. <https://doi.org/10.1038/nature11982>
- Fryirs, K. (2013). (dis)connectivity in catchment sediment cascades: A fresh look at the sediment delivery problem: (DIS)CONNECTIVITY in catchment sediment cascades. *Earth Surface Processes and Landforms*, 38(1), 30–46. <https://doi.org/10.1002/esp.3242>
- Godin, E., & Fortier, D. (2012). Geomorphology of a thermo-erosion gully, bylot island, nunavut, Canada ¹ this article is one of a series of papers published in this CJES special issue on the theme of *fundamental and applied research on permafrost in Canada*. ² polar continental shelf project contribution 043-11. *Canadian Journal of Earth Sciences*, 49(8), 979–986. <https://doi.org/10.1139/e2012-015>
- Harvey, A. M. (2001). Coupling between hillslopes and channels in upland fluvial systems: Implications for landscape sensitivity, illustrated from the howgill fells, northwest England. *Catena (Cremlingen)*, 42(2–4), 225–250. [https://doi.org/10.1016/S0341-8162\(00\)00139-9](https://doi.org/10.1016/S0341-8162(00)00139-9)
- Heckmann, T., Cavalli, M., Cerdan, O., Foerster, S., Javaux, M., Lode, E., et al. (2018). Indices of sediment connectivity: Opportunities, challenges and limitations. *Earth-Science Reviews*, 187, 77–108. <https://doi.org/10.1016/j.earscirev.2018.08.004>

- Hilger, L., & Beylich, A. A. (2019). Sediment budgets in high-mountain areas: Review and challenges. In T. Heckmann & D. Morche (Eds.), *Geomorphology of Proglacial Systems* (pp. 251–269). Springer International Publishing. https://doi.org/10.1007/978-3-319-94184-4_15
- Hooke, J., & Souza, J. (2021). Challenges of mapping, modelling and quantifying sediment connectivity. *Earth-Science Reviews*, 223, 103847. <https://doi.org/10.1016/j.earscirev.2021.103847>
- Hugonnet, R., McNabb, R., Berthier, E., Menounos, B., Nuth, C., Girod, L., et al. (2021). Accelerated global glacier mass loss in the early twenty-first century. *Nature*, 592(7856), 726–731. <https://doi.org/10.1038/s41586-021-03436-z>
- Husic, A., & Michalek, A. (2022). Structural hillslope connectivity is driven by tectonics more than climate and modulates hydrologic extremes and benefits. *Geophysical Research Letters*, 49(15), e2022GL099898. <https://doi.org/10.1029/2022GL099898>
- Huss, M., & Hock, R. (2018). Global-scale hydrological response to future glacier mass loss. *Nature Climate Change*, 8(2), 135–140. <https://doi.org/10.1038/s41558-017-0049-x>
- Immerzeel, W. W., Lutz, A. F., Andrade, M., Bahl, A., Biemans, H., Bolch, T., et al. (2020). Importance and vulnerability of the world's water towers. *Nature*, 577(7790), 364–369. <https://doi.org/10.1038/s41586-019-1822-y>
- Kirschbaum, D., Kapnick, S. B., Stanley, T., & Pascale, S. (2020). Changes in extreme precipitation and landslides over High Mountain Asia. *Geophysical Research Letters*, 47(4), e2019GL085347. <https://doi.org/10.1029/2019GL085347>
- Kokelj, S. V., Lacelle, D., Lantz, T. C., Tunnicliffe, J., Malone, L., Clark, I. D., & Chin, K. S. (2013). Thawing of massive ground ice in mega slumps drives increases in stream sediment and solute flux across a range of watershed scales. *Journal of Geophysical Research: Earth Surface*, 118(2), 681–692. <https://doi.org/10.1002/jgrf.20063>
- Kondolf, G. M., Gao, Y., Annandale, G. W., Morris, G. L., Jiang, E., Zhang, J., et al. (2014). Sustainable sediment management in reservoirs and regulated rivers: Experiences from five continents. *Earth's Future*, 2(5), 256–280. <https://doi.org/10.1002/2013EF000184>
- Laboratory (JPL). (2021). NASADEM merged DEM global 1 arc second V001 [Dataset]. <https://doi.org/10.5069/G93T9FD9>
- Lane, S. N., Bakker, M., Gabbud, C., Micheletti, N., & Saugy, J.-N. (2017). Sediment export, transient landscape response and catchment-scale connectivity following rapid climate warming and Alpine glacier recession. *Geomorphology*, 277, 210–227. <https://doi.org/10.1016/j.geomorph.2016.02.015>
- Lane, S. N., Reaney, S. M., & Heathwaite, A. L. (2009). Representation of landscape hydrological connectivity using a topographically driven surface flow index. *Water Resources Research*, 45(8), 2008WR007336. <https://doi.org/10.1029/2008WR007336>
- Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems [Dataset]. *Hydrological Processes*, 27(15), 2171–2186. <https://doi.org/10.1002/hyp.9740>
- Lewkowicz, A. G., & Way, R. G. (2019). Extremes of summer climate trigger thousands of thermokarst landslides in a high arctic environment. *Nature Communications*, 10(1), 1329. <https://doi.org/10.1038/s41467-019-09314-7>
- Li, D., Li, Z., Zhou, Y., & Lu, X. (2020). Substantial increases in the water and sediment fluxes in the headwater region of the Tibetan plateau in response to global warming. *Geophysical Research Letters*, 47(11), e2020GL087745. <https://doi.org/10.1029/2020GL087745>
- Li, D., Lu, X., Overeem, I., Walling, D. E., Syvitski, J., Kettner, A. J., et al. (2021). Exceptional increases in fluvial sediment fluxes in a warmer and wetter High Mountain Asia. *Science*, 374(6567), 599–603. <https://doi.org/10.1126/science.abi9649>
- Li, D., Lu, X., Walling, D. E., Zhang, T., Steiner, J. F., Wasson, R. J., et al. (2022). High Mountain Asia hydropower systems threatened by climate-driven landscape instability. *Nature Geoscience*, 15(7), 520–530. <https://doi.org/10.1038/s41561-022-00953-y>
- Li, D., Overeem, I., Kettner, A. J., Zhou, Y., & Lu, X. (2021). Air temperature regulates erodible landscape, water, and sediment fluxes in the permafrost-dominated catchment on the Tibetan Plateau. *Water Resources Research*, 57(2), e2020WR028193. <https://doi.org/10.1029/2020WR028193>
- Li, D., Zhang, T., Walling, D. E., Lane, S., Bookhagen, B., Tian, S., et al. (2024). The competing controls of glaciers, precipitation, and vegetation on high-mountain fluvial sediment yields. *Science Advances*, 10(48), eads6196. <https://doi.org/10.1126/sciadv.ads6196>
- Li, H., Mao, D., Li, X., Wang, Z., & Wang, C. (2019). Monitoring 40-year lake area changes of the qaidam basin, Tibetan plateau, using landsat time series. *Remote Sensing*, 11(3), 343. <https://doi.org/10.3390/rs11030343>
- Li, J., Wang, G., Li, K., Li, Y., Guo, L., & Song, C. (2023). Impacts of climate change and freeze–thaw cycles on water and sediment fluxes in the headwater region of the Yangtze River, qinghai–tibet plateau. *Catena*, 227, 107112. <https://doi.org/10.1016/j.catena.2023.107112>
- Li, J., Wang, G., Song, C., Sun, S., Ma, J., Wang, Y., et al. (2024). Recent intensified erosion and massive sediment deposition in Tibetan Plateau rivers. *Nature Communications*, 15(1), 722. <https://doi.org/10.1038/s41467-024-44982-0>
- Li, X., Long, D., Slater, L. J., Moulds, S., Shahid, M., Han, P., & Zhao, F. (2023). Soil moisture to runoff (SM2R): A data-driven model for runoff estimation across poorly gauged Asian water towers based on soil moisture dynamics. *Water Resources Research*, 59(3), e2022WR033597. <https://doi.org/10.1029/2022WR033597>
- Li, Z., Lu, H., Gao, P., You, Y., & Hu, X. (2020). Characterizing braided rivers in two nested watersheds in the source region of the Yangtze River on the qinghai–tibet plateau. *Geomorphology*, 351, 106945. <https://doi.org/10.1016/j.geomorph.2019.106945>
- Liu, C., Chen, Y., Fang, G., Li, Z., & Liu, Y. (2024). Impact of climatic and geomorphologic drivers on sediment connectivity in the tarim river basin, China. *Journal of Hydrology*, 643, 132027. <https://doi.org/10.1016/j.jhydrol.2024.132027>
- Livingstone, S. J., Utting, D. J., Ruffell, A., Clark, C. D., Pawley, S., Atkinson, N., & Fowler, A. C. (2016). Discovery of relict subglacial lakes and their geometry and mechanism of drainage. *Nature Communications*, 7(1), ncomms11767. <https://doi.org/10.1038/ncomms11767>
- López-Vicente, M., & Ben-Salem, N. (2019). Computing structural and functional flow and sediment connectivity with a new aggregated index: A case study in a large mediterranean catchment. *Science of the Total Environment*, 651, 179–191. <https://doi.org/10.1016/j.scitotenv.2018.09.170>
- Luo, J., Niu, F., Lin, Z., Liu, M., Yin, G., & Gao, Z. (2022). Inventory and frequency of retrogressive thaw slumps in permafrost region of the Qinghai–Tibet Plateau. *Geophysical Research Letters*, 49(23), e2022GL099829. <https://doi.org/10.1029/2022GL099829>
- Ma, J., Wang, G., & Li, H. (2024). A variance-upscaling quantile mapping method for gridded precipitation bias correction. *Journal of Hydrology*, 632, 130959. <https://doi.org/10.1016/j.jhydrol.2024.130959>
- Ma, J., Wang, G., Sun, S., Song, C., Li, J., Guo, L., et al. (2025). An enhanced image stacks method for mapping long-term retrogressive thaw slumps in the Tibetan Plateau. *International Journal of Applied Earth Observation and Geoinformation*, 137, 104414. <https://doi.org/10.1016/j.jag.2025.104414>
- Mancini, D., & Lane, S. N. (2020). Changes in sediment connectivity following glacial debuttreasing in an alpine valley system. *Geomorphology*, 352, 106987. <https://doi.org/10.1016/j.geomorph.2019.106987>
- Mayor, Á. G., Bautista, S., Small, E. E., Dixon, M., & Bellot, J. (2008). Measurement of the connectivity of runoff source areas as determined by vegetation pattern and topography: A tool for assessing potential water and soil losses in drylands. *Water Resources Research*, 44(10), 2007WR006367. <https://doi.org/10.1029/2007WR006367>
- McVey, I., Michalek, A., Mahoney, T., & Husic, A. (2023). Urbanization as a limiter and catalyst of watershed-scale sediment transport: Insights from probabilistic connectivity modeling. *Science of The Total Environment*, 894, 165093. <https://doi.org/10.1016/j.scitotenv.2023.165093>

- Messenzehl, K., Hoffmann, T., & Dikau, R. (2014). Sediment connectivity in the high-alpine valley of val Müschauns, Swiss national Park — Linking geomorphic field mapping with geomorphometric modelling. *Geomorphology*, 221, 215–229. <https://doi.org/10.1016/j.geomorph.2014.05.033>
- Michalek, A. T., Villarini, G., & Husic, A. (2023). Climate change projected to impact structural hillslope connectivity at the global scale. *Nature Communications*, 14(1), 6788. <https://doi.org/10.1038/s41467-023-42384-2>
- Mishra, K., Sinha, R., Jain, V., Nepal, S., & Uddin, K. (2019). Towards the assessment of sediment connectivity in a large Himalayan river basin. *Science of The Total Environment*, 661, 251–265. <https://doi.org/10.1016/j.scitotenv.2019.01.118>
- Mountain Research Initiative EDW Working Group. (2015). Elevation-dependent warming in mountain regions of the world. *Nature Climate Change*, 5(5), 424–430. <https://doi.org/10.1038/nclimate2563>
- Najafi, S., Dragovich, D., Heckmann, T., & Sadeghi, S. H. (2021). Sediment connectivity concepts and approaches. *Catena (Cremlingen)*, 196, 104880. <https://doi.org/10.1016/j.catena.2020.104880>
- Norris, S. L., Margold, M., Evans, D. J. A., Atkinson, N., & Froese, D. G. (2024). Dynamical response of the southwestern laurentide ice sheet to rapid boiling–allered warming. *The Cryosphere*, 18(4), 1533–1559. <https://doi.org/10.5194/tc-18-1533-2024>
- Ortiz-Rodríguez, A. J., Muñoz-Robles, C., & Borselli, L. (2019). Changes in connectivity and hydrological efficiency following wildland fires in sierra madre oriental, Mexico. *Science of the Total Environment*, 655, 112–128. <https://doi.org/10.1016/j.scitotenv.2018.11.236>
- Overeem, I., Hudson, B. D., Syvitski, J. P. M., Mikkelsen, A. B., Hasholt, B., Van Den Broeke, M. R., et al. (2017). Substantial export of suspended sediment to the global oceans from glacial erosion in Greenland. *Nature Geoscience*, 10(11), 859–863. <https://doi.org/10.1038/ngeo3046>
- Persichillo, M. G., Bordon, M., Cavalli, M., Crema, S., & Meisina, C. (2018). The role of human activities on sediment connectivity of shallow landslides. *Catena (Cremlingen)*, 160, 261–274. <https://doi.org/10.1016/j.catena.2017.09.025>
- Restrepo, J. D., & Kjerfve, B. (2000). Magdalena river: Interannual variability (1975–1995) and revised water discharge and sediment load estimates. *Journal of Hydrology*, 235(1), 137–149. [https://doi.org/10.1016/S0022-1694\(00\)00269-9](https://doi.org/10.1016/S0022-1694(00)00269-9)
- Rodzik, J., Furtak, T., & Zglobicki, W. (2009). The impact of snowmelt and heavy rainfall runoff on erosion rates in a gully system, Lublin Upland, Poland. *Earth Surface Processes and Landforms*, 34(14), 1938–1950. <https://doi.org/10.1002/esp.1882>
- Rudy, A. C. A., Lamoureux, S. F., Kokelj, S. V., Smith, I. R., & England, J. H. (2017). Accelerating thermokarst transforms ice-cored terrain triggering a downstream cascade to the ocean. *Geophysical Research Letters*, 44(21), 11080–11087. <https://doi.org/10.1002/2017GL074912>
- Seybold, H., Berghuijs, W. R., Prancevic, J. P., & Kirchner, J. W. (2021). Global dominance of tectonics over climate in shaping river longitudinal profiles. *Nature Geoscience*, 14(7), 503–507. <https://doi.org/10.1038/s41561-021-00720-5>
- Shen, C., Jia, L., & Ren, S. (2022). Inter- and intra-annual glacier elevation change in High Mountain Asia region based on ICESat-1 and 2 data using elevation-aspect bin analysis method. *Remote Sensing*, 14(7), 1630. <https://doi.org/10.3390/rs14071630>
- Shi, X., Zhang, F., Lu, X., Wang, Z., Gong, T., Wang, G., & Zhang, H. (2018). Spatiotemporal variations of suspended sediment transport in the upstream and midstream of the Yarlung Tsangpo River (the upper Brahmaputra), China. *Earth Surface Processes and Landforms*, 43(2), 432–443. <https://doi.org/10.1002/esp.4258>
- Syvitski, J. P. M. (2002). Sediment discharge variability in arctic rivers: Implications for a warmer future. *Polar Research*, 21(2), 323–330. <https://doi.org/10.3402/polar.v21i2.6494>
- Tang, M., Xu, Q., Wang, L., Zhao, H., Wu, G., Zhou, J., et al. (2023). Hidden dangers of ice avalanches and glacier lake outburst floods on the Tibetan plateau: Identification, inventory, and distribution. *Landslides*, 20(12), 2563–2581. <https://doi.org/10.1007/s10346-023-02125-4>
- Tang, W., Zhou, J., Ma, J., Wang, Z., Ding, L., Zhang, X., & Zhang, X. (2024). Trims LST: A daily 1 km all-weather land surface temperature dataset for China's landmass and surrounding areas (2000–2022). *Earth System Science Data*, 16(1), 387–419. <https://doi.org/10.5194/essd-16-387-2024>
- Tarboton, D. G. (1997). A new method for the determination of flow directions and upslope areas in grid digital elevation models. *Water Resources Research*, 33(2), 309–319. <https://doi.org/10.1029/96WR03137>
- Tucker, G. E., & Slingerland, R. (1997). Drainage basin responses to climate change. *Water Resources Research*, 33(8), 2031–2047. <https://doi.org/10.1029/97WR00409>
- Wang, G., Hu, H., & Li, T. (2009). The influence of freeze–thaw cycles of active soil layer on surface runoff in a permafrost watershed. *Journal of Hydrology*, 375(3–4), 438–449. <https://doi.org/10.1016/j.jhydrol.2009.06.046>
- Wang, H., Saito, Y., Zhang, Y., Bi, N., Sun, X., & Yang, Z. (2011). Recent changes of sediment flux to the western Pacific Ocean from major rivers in East and Southeast Asia. *Earth-Science Reviews*, 108(1–2), 80–100. <https://doi.org/10.1016/j.earscirev.2011.06.003>
- Wang, H., Xu, C., Ying, G., Liu, F., & Long, Y. (2022). Variations in glacier runoff contributed by the increased negative mass balance over the last forty years in the tien Shan mountains. *Water*, 14(7), 1006. <https://doi.org/10.3390/w14071006>
- Wang, L., MacLennan, S. A., & Cheng, F. (2020). From a proximal-deposition-dominated basin sink to a significant sediment source to the Chinese loess plateau: Insight from the quantitative provenance analysis on the cenozoic sediments in the qaidam basin, northern Tibetan plateau. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 556, 109883. <https://doi.org/10.1016/j.palaeo.2020.109883>
- Wang, L., Yao, T., Chai, C., Cuo, L., Su, F., Zhang, F., et al. (2021). TP-River: Monitoring and quantifying total river runoff from the third Pole. *Bulletin of the American Meteorological Society*, 102(5), E948–E965. <https://doi.org/10.1175/BAMS-D-20-0207.1>
- Wasson, R., Acharjee, S., & Rakshit, R. (2022). Towards identification of sediment sources, and processes of sediment production, in the Yarlung-Tsangpo-Brahmaputra River catchment for reduction of fluvial sediment loads. *Earth-Science Reviews*, 226, 103932. <https://doi.org/10.1016/j.earscirev.2022.103932>
- Wohl, E., Brierley, G., Cadol, D., Coulthard, T. J., Covino, T., Fryirs, K. A., et al. (2019). Connectivity as an emergent property of geomorphic systems. *Earth Surface Processes and Landforms*, 44(1), 4–26. <https://doi.org/10.1002/esp.4434>
- Wu, Z., Baartman, J. E. M., Pedro Nunes, J., & López-Vicente, M. (2023). Intra-annual sediment dynamic assessment in the wei river basin, China, using the AIC functional-structural connectivity index. *Ecological Indicators*, 146, 109775. <https://doi.org/10.1016/j.ecolind.2022.109775>
- Yang, G., Qiu, H., Wang, N., Yang, D., & Liu, Y. (2025). Tracking 35-year dynamics of retrogressive thaw slumps across permafrost regions of the Tibetan Plateau. *Remote Sensing of Environment*, 325, 114786. <https://doi.org/10.1016/j.rse.2025.114786>
- Yao, T., Thompson, L., Yang, W., Yu, W., Gao, Y., Guo, X., et al. (2012). Different glacier status with atmospheric circulations in Tibetan Plateau and surroundings. *Nature Climate Change*, 2(9), 663–667. <https://doi.org/10.1038/nclimate1580>
- Zhang, F., Zeng, C., Wang, G., Wang, L., & Shi, X. (2022). Runoff and sediment yield in relation to precipitation, temperature and glaciers on the Tibetan Plateau. *International Soil and Water Conservation Research*, 10(2), 197–207. <https://doi.org/10.1016/j.iswcr.2021.09.004>
- Zhang, J., Xu, B., Gu, Z., Lv, Y., Yin, Z., Guo, X., & Li, L. (2023). Coupling of river discharges and alpine glaciers in arid central asia. *Quaternary International*, 667, 19–28. <https://doi.org/10.1016/j.quaint.2023.06.002>

- Zhang, T., Li, D., East, A. E., Kettner, A. J., Best, J., Ni, J., & Lu, X. (2023). Shifted sediment-transport regimes by climate change and amplified hydrological variability in cryosphere-fed rivers. *Science Advances*, 9(45), eadi5019. <https://doi.org/10.1126/sciadv.adi5019>
- Zhang, T., Li, D., East, A. E., Walling, D. E., Lane, S., Overeem, I., et al. (2022). Warming-driven erosion and sediment transport in cold regions. *Nature Reviews Earth and Environment*, 3(12), 832–851. <https://doi.org/10.1038/s43017-022-00362-0>
- Zhao, L., Zou, D., Hu, G., Wu, T., Du, E., Liu, G., et al. (2021). A synthesis dataset of permafrost thermal state for the qinghai–tibet (xizang) plateau, China. *Earth System Science Data*, 13(8), 4207–4218. <https://doi.org/10.5194/essd-13-4207-2021>